Existence of Solutions Algebraic Equations from Existence of Solutions of Ordinary Differential Equations

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Existence theorem of solutions of ordinary differential equations (Cauchy-Lipschitz theorem) follows without existence of algebraic eqations. We can proof Existence of solutions of algebraic equations from existence of solutions of ordinary differential equations.

Linear ordinary differential eqations with constant coefficients on bounded and closed interval I , an initial value probrem $P(D)u(x) = 0,\ x \in I, u^{(k)}(0) = u_{0k}\ (k = 0, \dots, n-1)$ has an unique solution $u \in C^1(\bar{J})$ from corollary of Cauchy-Lipschitz theorem ([1]): $0 = P(x) = \sum_{i=0}^{n} a_i x^i \Rightarrow x^n = Q(x) = \sum_{i=0}^{n-1} -(a_i/a_n)x^i$, $Q(x, x_1, \dots, x_{n-1}) = \sum_{i=0}^{n-1} -(a_i/a_n)x_i^i\ (x_0 = x)$ is Lipschitz continuous, so $u^{(n)}(x) = Q(D)u(x), u^{(k)}(0) = u_{0k}$ has a solution on the neighborhood $0 \in J = \bar{J} \subset I$. To proof it, put $v = u^{(n-1)}, f(x, u, \dots, u^{(n-1)}) = Q(D)v(x)$, use mathematical induction and Cauchy-Lipschitz theorem to $v'(x) = f(x, u, v), v(0) = u^{(n-1)}(0)$.
u has an extension $\bar{u} \in \mathcal{S}$. By Fourier transform, we obtain $P(i\xi)\mathcal{F}\bar{u}(\xi) = 0$.
$P(i\xi)\mathcal{F}\bar{u}(\xi) = 0 \iff P(D)\bar{u}(x) = 0$ ([2]). We can choose $\bar{u}$ as $\exists\xi, \mathcal{F}\bar{u}(\xi) \neq 0$, so we have an existence of ξ such that $P(i\xi) = 0$.

References: [1]石原茂-矢野健太郎, 解析学概論, 裳華房, 2020; [2]垣田高雄-柴田良弘, ベクトル解析から流体へ, 日本評論社, 2024.